Quiz1 **Introduction to Programming** **Fall 2024**

School of Mathematics & Computer Science, IBA Karachi

ERP No.______________________

Instructions:

1. **Attempt all questions . Understanding question is part of quiz.**
2. **Calculators are not allowed**
3. **write the final answer in the space provided for it**

General Hints:

1. **Comparison of double/float type variables is exactly similar to int type variables**
2. **For arithmetic operators if one of the variable/value type is double/float and one is int then the answer is floating point value.**

SHORT QUESTIONS

1. The following program segment is designed to print a message if the input string is between the length of 5 and 10. What condition should be written within the parenthesis so that the program segment works correctly? Please remember that you are not allowed to use any other variables or add any new instruction in the program segment. Just write the condition using the already declared variable(s). [3 Points]

```
std::string message;
std::cin >> message;
if(______________________________________)
std::cout<< ______________
else
std::cout<< " Invalid Input";
```

Commented [MK1]: message.length()>5 && message.length()<5

Commented [MK2]: message

2. For each of the following program segments, assume that the code is contained within the main function and appropriate headers are also declared. Specify the output that each segment will produce when executed. If there is an error in any of the code, identify and describe the error.

[10 = (1 + 1 + 1 + 1 + 1 + 1 + 1 + 1+1+1+1+1) Points]

Program Segment	Output

int a = 10, b = 18;
if (a = b) std::cout << "equal";
else std::cout << "not equal";
	equal

#include <iostream>
using namespace std;
int main() {
  int x = 8;
  int y = 12;
  int result = --x + y-- + ++x + --y;
  cout << "x: " << x << endl;
  cout << "y: " << y << endl;
	x: 8
	y: 10
	result: 37

<table>
<tr><td><pre>   cout << "result: " << result << endl;
    return 0;}</pre></td><td></td></tr>
<tr><td><pre>#include <iostream>
using namespace std;
int main(){
int a = 17, x = 5, y = 12;
if (x > y);
{
a = 13;
x = 23;
}
std::cout << a;
}</pre></td><td>13</td></tr>
<tr><td><pre>int total = 1, z = 6;
if (z % 3 == 0)
    total *= z; z -= 2;
if (z % 4 == 0)
    total += z; z += 3;
if (z % 5 == 0)
    total -= z;
cout << total;</pre></td><td>10</td></tr>
<tr><td><pre>int a = 3, b = 4;
    switch (a) {
        case 2: cout << "alpha";
        case 3: cout << "beta";
        case 4: cout << "gamma";
        default: cout << "delta";
    }</pre></td><td>betagammadelta</td></tr>
<tr><td><pre>int a, b, c;
a = 4; b = 9; c = 2;
    if (a + b > c) {
        if (b - a < c)
            c = a;
        else
            a = b;
    }
    else {
        if (c * a > b)
            b = c;
        else
            a = c;
    }
 cout << "a = " << a << " b = " << b << " c = " << c << endl;</pre></td><td>a = 9 b = 9 c = 2</td></tr>
</table>

int x = 7, y = 14, z = 3; cout << ((x + z) < y) << ((y % x) == z) << (x * z > y);	101
int a = 30, b = 20, c = 10, d = 8; cout << (a + b / d – c) * (a % b) – (d / c);	220
int RollNo, rollno; RollNo = 5; rollno = 8; cout << "Roll No is " << rollno;	8

int cout = 5;
int result = ++cout + cout++;
std::cout << "Result: " << result << endl;
std::cout << "Final value of cout: " << std::cout << endl;
	Error

Commented [MK3]: This should be only cout as it is variable

3. The following program has been written to compute and display percentage of marks obtained. However, the code is not producing the desired result.

   Suggest a simple change in a single line of the program that will correct this logical error.

   You must also rewrite the corrected line (only) in the space provided below [3 Points]

```
#include <iostream>
using namespace std;

int main() {
    int total_marks = 450, obtained_marks = 300;
    double percentage;

    percentage = obtained_marks / total_marks * 100;

    cout << "Percentage: " << percentage << "%";
    return 0;
}
```

YOUR ANSWER GOES HERE

```
 percentage = ((double)obtained_marks / total_marks) * 100;
```

4. The following program has some syntax errors. Identify all the syntax errors and write each line correctly so that the program compiles without errors   [6 Points]

PROGRAM LINES WITH ERROERS	PROGRAM WITHOUT ERROR
#inclde<iostream>	#include <iostream>

Using namespace std;	using namespace std;
int main()	int main()
{	{
integer TC , FH;	int TC,FH;
cout << ‘Temperature IN’;	cout << “Temperature IN”;
cin << T;	cin>>TC;
F << 9\5.0* T + 32	FH=9\5.0* T + 32;
COUT >> “Temperature OUT” << F <Endl	cout << “Temperature OUT” << FH<endl;
Return 0;	return 0
}	}

Coding

Problem No 1: [Simple Interest Calculation] [6 Points]

Borrowers and lenders often need to calculate simple interest on loans or investments to understand how much interest will be paid or earned over time. For example, someone might want to know how much interest they will earn on a savings account over a year.

In this problem, you are required to write a simple program that will take three inputs:

i) the principal amount (the initial amount of money)

ii) the interest rate (as a percentage)

iii) the time period (in years),

and then display the total interest earned or paid. The formula for calculating simple interest is:

Simple Interest = Principal Amount × Interest Rate × Time Period

Write a program in the space provided below where the marks will be awarded for:

- Using the correct type of variables. [1 Point]
- Correct inputs and input validation (principal amount, interest rate, and time period > 0). [2 Points]
- Correct calculations. [2 Points]
- Correctly displaying the results and messages. [1 Point]

#write you code in space provided here

```
#include <iostream>
using namespace std;

int main() {
    double principal, interestRate;
    int timePeriod;

    cout << "Enter the principal amount: ";
    cin >> principal;

    cout << "Enter the interest rate (as a percentage): ";
    cin >> interestRate;

    if (interestRate <= 0) {
        cout << "Interest rate must be greater than 0." << endl;
        return 1;
    }

    cout << "Enter the time period (in years): ";
    cin >> timePeriod;

    if (timePeriod <= 0) {
        cout << "Time period must be greater than 0." << endl;
        return 1;
    }

    double simpleInterest = (principal * interestRate * timePeriod) / 100;

    cout << "The total interest earned or paid is: " << simpleInterest << endl;

    return 0;
}
```

Commented [MK4]: Both int and double are acceptable depending on assumption as year can also be 1.5

Problem No 2: [Caloric Needs Calculator] [12 Points]

Caloric needs for an individual are calculated based on their weight (in kilograms), height (in centimeters), age (in years), and activity level. The Basal Metabolic Rate (BMR) can be calculated using the following formulas:

For Men:

BMR = 88.362 + (13.397 × weight) + (4.799 × height) - (5.677 × age)

For Women:

BMR = 447.593 + (9.247 × weight) + (3.098 × height) - (4.330 × age)

To calculate daily caloric needs, multiply BMR by the activity factor:

- Sedentary (little or no exercise): BMR × 1.2
- Lightly active (light exercise/sports 1-3 days/week): BMR × 1.375
- Moderately active (moderate exercise/sports 3-5 days/week): BMR × 1.55
- Very active (hard exercise/sports 6-7 days a week): BMR × 1.725
- Super active (very hard exercise, a physical job, or training twice a day): BMR × 1.9

Write a program that inputs the user's gender, weight in kilograms, height in centimeters, age in years, and activity level. The program should calculate and display the user's daily caloric needs. Marks will be awarded for:

- Correct indentation. [2 Points]
- Using a sufficient number of and correct type of variables. [2 Points]
- Correct conversions and calculations. [4 Points]
- Correctly displaying the answers and messages. [5 Points]

Write your program in the space provided on the next page.

```
#include <iostream>
#include <string>
using namespace std;

int main() {
    string gender;
    double weight, height, age;
    int activityLevel;
    double bmr, dailyCalories;

    cout << "Enter your gender (male/female): ";
    cin >> gender;

    cout << "Enter your weight (in kilograms): ";
    cin >> weight;

    cout << "Enter your height (in centimeters): ";
    cin >> height;

    cout << "Enter your age (in years): ";
    cin >> age;

    cout << "Enter your activity level (1 for Sedentary, 2 for Lightly active, 3 for Moderately active, 4 for
Very active, 5 for Super active): ";
    cin >> activityLevel;

    if (gender == "male") {
        bmr = 88.362 + (13.397 * weight) + (4.799 * height) - (5.677 * age);
    } else if (gender == "female") {
        bmr = 447.593 + (9.247 * weight) + (3.098 * height) - (4.330 * age);
    } else {
        cout << "Invalid gender entered." << endl;
        return 1;
    }

    switch (activityLevel) {
        case 1:
            dailyCalories = bmr * 1.2;
            break;
        case 2:
            dailyCalories = bmr * 1.375;
            break;
        case 3:
            dailyCalories = bmr * 1.55;
            break;
        case 4:
            dailyCalories = bmr * 1.725;
            break;
```

```
    case 5:
        dailyCalories = bmr * 1.9;
        break;
    default:
        cout << "Invalid activity level entered." << endl;
        return 1;
    }

    cout << "Your daily caloric needs are: " << dailyCalories << " calories." << endl;

    return 0;
}
```

Commented [MK5]: Solution provided uses switch case but you can also use if/else

Problem No 3: Make changes in the following program so that it uses a switch-case statement in place of the if-else statement [10 points]

```
 #include <iostream>
using namespace std;
int main()
{
int noOfWheels=0;
cout << "Enter number of wheels  ";
cin >> noOfWheels;
if(noOfWheels == 1) cout <<"Unicycle";
else if(noOfWheels == 2) cout << "Motorcycle";
else if(noOfWheels == 3) cout << "Rikshaw";
else if(noOfWheels == 4) cout << "Car";
else cout << " Heavy loaded vehicles ";}
```

```
#write your code here.

 #include <iostream>
using namespace std;

int main() {
    int noOfWheels = 0;
    cout << "Enter number of wheels: ";
    cin >> noOfWheels;

    switch (noOfWheels) {
        case 1:
            cout << "Unicycle";
            break;
        case 2:
            cout << "Motorcycle";
            break;
        case 3:
            cout << "Rikshaw";
            break;
        case 4:
            cout << "Car";
            break;
        default:
            cout << "Heavy loaded vehicles";
            break;
    }

    return 0;
}
```